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An Overview of Holistic Face Recognition

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***Abstract*—Biometric is including two parts bios (life) and metrikos (measure). This word comes from Greek. A biometric system is essentially a pattern recognition system that recognizes a person based on a feature vector derived from a specific physiological or behavioral characteristic that the person possesses. Holistic face recognition utilizes global information from faces to perform face recognition. The global information from faces is fundamentally represented by a small number of features, which are directly derived from the pixel information of face images. These small numbers of features distinctly capture the variance among different individual faces and therefore are used to uniquely identify individuals. Describe two holistic approaches to face recognition called Karhunen-Lo`eve expansion and Linear Discriminate Analysis.**

Key words, ***Biometric, holistic, Karhunen-Lo`eve expansion, Linear Discriminate Analysis.***

I. Introduction

All Biometric is including two parts bios (life) and metrikos (measure). This word comes from Greek. A biometric system is essentially a pattern recognition system that recognizes a person based on a feature vector derived from a specific physiological or behavioral characteristic that the person possesses [1]. That feature vector is usually stored in a database (or recorded on a smart card given to the individual) after being extracted.

II. Brief overview biometric technology about face

Facial images are the for the most part familiar biometric characteristic used by humans to make a personal recognition, hence the idea to use this biometric in technology. This is a nonintrusive method and is suitable for covert recognition applications. The applications of facial recognition range from static ("mug shots") to dynamic, uncontrolled face identification in a cluttered background (subway, airport) [15]. Face verification involves extracting a feature set from a two-dimensional image of the user's face and matching it with the template stored in a database. The most popular approaches to face recognition are based on either:

1) The location and shape of facial attributes such as eyes, eyebrows, nose, lips and chin, and their spatial relationships.

2) The overall (global) analysis of the face image that represents a face as a weighted combination of a number of canonical faces [2]. Although performance of commercially available systems is reasonable there is still significant room for improvement since false reject rate (FRR) is about 10% and false accept rate (FAR) is 1% [3]. These systems also have difficulties in recognizing a face from images captured from two different angles and under different ambient illumination conditions. It is questionable if a face itself is a sufficient basis for recognizing a person from a large number of identities with an extremely high level of confidence [4]. Facial recognition system should be able to automatically detect a face in an image, extract its features and then recognize it from a general viewpoint (is., from any pose) which is a rather difficult task. Another problem is the fact that the face is a changeable social organ displaying a variety of expressions.

III. Face recognition

The More than 20 years has been Research on face recognition technology. Face recognition has become most important biometrics authentication technology in the past a few years .It has become a new major research area in the last few years because of the number of potential applications. These applications range from security access control, personal identification to human-computer communication [15,40]. A number of face recognition methods have been proposed. These methods can be divided into two approaches: namely, constituent and face based. In the constituent based approach, recognition is based on the relationship between human facial features such as eye(s), mouth, nose, profile silhouettes and face boundary [6,5].

Deformable template is a common method used in detecting facial features. First proposed using deformable template for eye detection in 1992.Direction [6] modified the eye template and the energy function for eye detection. Okada et al. [7] proposed to use Gabor wavelet to extract facial features, called JETS. ”Ref [5]” proposed the concept of eye corners to guide the template matching process. “Ref [10]” proposed integration projections to extract the facial feature which is an extension of Kanade's work [10].”Ref [10]” used chrominance information to locate face and facial features. Once facial features are detected, classification can be performed using the inter-feature distance [11], Dempster-Shafer theory [12], random graph [13] or elastic graph matching technique [14]. Constituent-based approach provides high flexibility in handling non-rigid facial features, such as eyes and mouth. On the other hand, the success of this

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approach highly relies on the accuracy of the facial feature detection scheme. Moreover, as mentioned by [15, 16], the image resolution has to be relatively high in order to detect the facial features accurately. In turn, the computational load will be increased. Face-based approach [17-22] attempts to capture and define the face as a whole. The face is treated as a two-dimensional pattern of intensity variation. Under this approach, the face is matched through finding its under-lying statistical regularities. Principal component analysis (PCA) is a typical face-based technique. "Ref [17,23]"first proposed to use Karhunen- Loe`ve transform to represent human faces. In their method, faces are represented by a linear combination of weighted eigenvector. "Ref [18]" developed a face recognition system using PCA (K-L expansion). Along this direction, many PCA-based recognition sys-tem has been developed [19,24]. Theoretically, face-based (PCA-based) approach has a number of practical advantages. First, as this approach considers face as a global feature, detection of local feature is not required. Second, the resolution of the operating image can be relatively low, which will reduce the computational load. However, image registration is required in PCA-based approach. This approach gives a very good result if the face to be recognized is aligned properly [22,41]. Connectionist method [25] also belongs to this approach. In [20] developed an analytic-to-holistic approach for frontal-view face recognition. Fast Fourier Transform was another approach applied to face recognition. [5] They proposed using the Fast Fourier Transform approach to represent the gallery set as a number of templates. Face recognition for this approach was achieved by minimizing the Euclidean distance amongst the peaks found between comparing a gallery face to a probe face. Another face recognition approach was the Hidden Markov Model. This approach had been used in speech and character recognition, but was applied to face recognition by [26].

The mean of face recognition is to identify or verify one or more persons from still images or video images of a scene using a stored database of faces. Many research efforts [27] have been focused on how to improve the accuracy of a recognition system, however, it seems that most of them ignore the potential problem that may stem from the face database at hand, where there may be only one sample image per person stored, possibly due to the difficulties of collecting samples or the limitations of storage capability of the systems, etc. As mentioned before in the face recognition a lot of, techniques used [29].

IV. Holistic methods

Holistic face recognition utilizes global information from faces to perform face recognition. The global information from faces is fundamentally represented by a small number of features, which are directly derived from the pixel information of face images. These small numbers of features distinctly capture the variance among different individual faces and therefore are used to uniquely identify individuals. Holistic methods use the whole face as input. In these methods, each face image is represented as signal high-dimensional vector by concatenating the gray values of all pixels in the face. These methods identify a face using the whole face image as input. There are four categorizations of holistic methods are as follows:

A. Karhunen-Le eve expansion

Use Expansion concept and explain the use of partial and full faces by [25] as well as, an influential paper on the Eigen face approach by [26] Additionally, we will also discuss the Karhunen-Lo`eve expansion been applied to face recognition by others . The Karhunen-Lo`eve expansion, also known as Principal Component Analysis or Hoteling transform, is traditionally concerned with feature selection for signal representation. Extracting small number of feature that identified by principle component of the face when used karhunen-lo eve to face recognition. The principal components of the face are found by projecting two dimensional face images into a one-dimensional subspace, then selecting the principal components, which capture the highest variances amongst individual faces.

B. Linear Discriminate Analysis (LDA)

The Fisher's Linear Discriminate also called Linear Discriminate Analysis discovers few features to distinguish faces of different or same individual. Few features is found by maximizing the Fisher Discriminate Criterion (Fisher 1936), which is achieved by maximizing the grouping of individual faces whilst minimizing the grouping of different individual faces. Therefore by grouping faces of the same individual these features can be used to determine the identity of individuals [8].

C. principal component analysis (PCA)

The principal component analysis (PCA) is a kind of algorithms in biometrics. It is a statistics technical and used or-thogonal transformation to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables. PCA also is a tool to reduce multidimensional data to lower dimensions while retaining most of the information [39,19].

D.Noise Model

This model proposes for authentication for corrupted faced.
Noise model include two phases:

- Training phase: in this phase great corrupted image by controlling noise parameter.
- Testing phase: in this phase can estimate noise parameter with corrupted image and original image [37].

In this model can synthesize face image from the original face image with estimate noise parameters and verify it with corrupted face image [37].

V. Enlarge the size of training set

After Another way to solve the one sample problem using holistic feature is to enlarge the size of training set (labeled set of face imaging). This can be done by synthesizing virtual samples. Here virtual samples include both new representations for the single training sample and new visual samples that are not previously existed in the database. It follows then that for virtual samples generation, one can choose to either construct

new representations or create novel visual images, based on the sing training image. The fundamental difference between the two methods lies in whether prior information is used: the representation construction method focuses on mining more information from the face image at hand, while visual sample generation method concentrates on learning extra information from the domain, besides the given training set [38]. Note that, however, the boundary between the two methods is not so crisp, since novel visual image can also be thought of as a new representation of the given face. Both the two types of methods are discussed in more detail in the subsection below.

A. Construct new representations

To be success in recognition, representation is important in recognition system. Different type of noises and appearance change is used for different representation. Therefore, it is generally is very hard to select the optimal representation that maximizes recognition performance. For example, one can combine various representations of the same image so as to exploit their specific merit. Motivated by this, [34] presented a method named representational oriented component analysis (ROCA), basing on one sample per person. In particular, each gallery image is firstly preprocessed to mitigate the effects of light direction changes, and then, several linear and nonlinear filters are applied on each image to produce its 150 representations in total. Next, an oriented component analysis (OCA) classifier is built on each representation. Finally, all the OCA classifiers are combined with a weighted linear sum method to give the final decision. Several numerical methods have been introduced to improve the OCA classifier's generalization [34]. As a conclusion, a large number of new sample are created for each person. In addition, it is used for standard Eigen space technique for feature extraction and recognition.

B. Generate novel views

Towards the needed of virtual samples should be diverse and representative enough, that is, they occupy different locations in the face space and represent specific variations of face images. One possible way to achieve the goal is to exploit prior information that can be learned from prototypical examples in the domain. As a simple way to create new samples, one can introduce some geometrical transformations on the original face images, such as rotation, scale transformation, and bilaterally symmetric transformation [35]. More complicated ways to create virtual samples with prior knowledge mainly aim to address the face recognition problem in uncontrolled conditions (outdoor environment), where pose and illumination may change significant. This technique may fail to work under the condition of partial occlusions or too large variation. A possible solution to this problem that reconstructs partially damaged face images from other faces was presented [36].

Here mentions about disadvantage of generate novel views:

- This method cannot recognize different transformation coefficients according to different transformation.
- For this method has difficult as the number of modes of variation grows large.
- Allow texture rendering for novel image, one need to know the correspondence between the real image and the referenced image.

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VI. Conclusion

In this paper, we are trying to explain about holistic face recognition and some methods related to biometric A. Karhunen-Le eve expansion, Noise, PCA and LDA methods are also presented.

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References

[1] A. K. Jain, A. Ross, S. Prabhakar, "An Introduction to Biometric Recognition",IEEE"FRVT2002:OverviewandSummary,"availab leat:http://www.frvt.orglFRVT2002/documents.h

[2] A. O'Toole, H. Abdi, K. De!enbacher, D. Valentin, Low-dimensional representation of faces in higher dimensions of the face space, J. Opt. Soc. Am. A. 10 (3) (1993) 405- 411.

[3] A.J. O'Toole, H. Abdi, Low-dimensional representation of faces in higher dimensions of the face space, Opt. Soc. Am. 10 (3) (1993) 405–411.

[4] A.L. Yuille, P.W. Hallinan, D.S. Cohen, Feature extraction from faces using deformable templates, Int. J Comput. Vision 8 (2) (1992) 99-111.

[5] B.W. Hwang, S.W. Lee, Reconstruction of partially damaged face images based on a morphable face model, IEEE Trans. Pattern Anal. Mach. Intell. 25 (3) (2003) 365–372.

[6] C.W. Chen, C.L. Huang, Human face recognition from a single front view, Int. J. Pattern Recognition Artif. Intel. 6 (4) (1993) 571-579.Concerns", IEEE Security & Privacy, MarcWApril2003, pp. 33-42.

[7] D. Maio, D. Maltoni, R. Cappelli, J. L. Wayman, A. K. Jain, "FVC2002: Fingerprint verification competition" in Proc. Int. Con$ Pattern Recognition (ICPR), Quebec City, QC, Canada, August 2002, pp. 744-747.

[8] D. S. Pankaj and M. Wilscy, "Comparison of PCA, LDA and Gabor Features for Face Recognition Using Fuzzy Neural Network," *in Advances in Computing and Information Technology: Springer*, 2013, pp. 413-422

[9] F. Frade, De la Torre, R. Gross, S. Baker, V. Kumar, Representational oriented component analysis (ROCA) for face recognition with one sample image per training class, in: Proceedings, IEEE Conference.

[10] G.C. Feng, P.C. Yuen, Variance projection function and its application to eye detection for human face recognition, Pattern Recognition Lett. 19 (1998) 899-906.

[11] H. Okada, J. Ste!ens, T. Maurer, H. Hong, E. Elagin, H. Neven, C. Malsburg, The Bochum/USC face recognition system, Face Recognition: From Theory to Applications, Springer, Berlin, 1998, pp. 186-205.

[12] H.C. Huang, M. Ouhyoung, J.L. Wu, Automatic feature point extraction on a human in model-based image cod-ing, Opt. Engng. 32 (7) (1993) 1571}1580.

[13] H.H.S. Ip, J.M.C. Ng, Human face recognition using Dempster-Shafer theory. Proceedings of IEEE Interna-tional Conference on Image Processing 1994, pp. 292-295.

[14] H.C. Jung, B.W. Hwang, S.W. Lee, Authenticating corrupted face image based on noise model, Proceedings of the Sixth IEEE International Conference on Automatic Face and Gesture Recognition, 2004, 272pp.

[15] J. Ashok, V. Shivashankar and P. Mudiraj, "An Overview of Biometrics," *International Journal,* Vol. 2, 2010.

[16] J. Weng, N. Ahuja, T.S. Huang, Learning recognition anegmentation of 3Dobjects from2D images, Proceedings of ICCV, 1993, pp. 121-128.

[17] J.H. Lai, P.C. Yuen, G.C. Feng, Spectroface: a Fourier-based approach for human face recognition, Proceeding of The second International Conference on Multimodal Interface, Hong Kong, 1999, pp. VI 115-120.

[18] J.Yang, D. Zhang, A.F. Frangi, J.Yang, Two-dimensional PCA: a new approach to appearance-based face representation and recognition, IEEE Trans. Pattern Anal. Mach. Intell. 26 (1) (2004) 131–137.

[19] J. Wu, Z.-H. Zhou, Face recognition with one training image per person, Pattern Recognition Lett. 23 (14) (2002) 1711–1719.

[20] K.M. Lam, H. Yan, An analytic-to-holistic approach for face recognition based on a single frontal view, IEEET rans. Pattern Anal. Mach. Intell. 20 (7) (1998) 673- 686.

[21] K.M. Lam, H. Yan, Location and extracting the eye inhuman face images, Pattern Recognition 29 (5) (1996) 771-779.

[22] K. S. Sodhi and M. Lal, "Face Recognition Using PCA, LDA and Various Distance Classifier," *Journal of Global Research in Computer Science,* Vol. 4, 2013, pp. 30-35.

[23] L. Sirovich, M. Kirby, Low-dimensional procedure for the characterization of human faces, J. Opt. Soc. Am. A. 4 (3)(1987) 519- 524.

[24] L. Sirovich, M. Kirby, Low-dimensional procedure for the characterization of human faces, J. Opt. Soc. Am. A 4 (3) (1987) 519–524.

[25] L. Wiskott, J.M. Fellous, N. Kruger, C. Malsburg, Facerecognition by elastic bunch graphmatching, IEEE Trans. Pattern Anal. Mach. Intell. 19 (7) (1997) 775-780.

[26] M. Golfarelli, D. Maio, D. Maltoni, "On the error-reject tradeoff in biometric verification systems," IEEE Trans. Pattern Anal. Machine Intell., Vol. 19, pp. 786-796, July 1997.

[27] M. Kirby, L. Sirovich, Application of karhunen-Loeve procedure for characterization of human faces, IEEE Trans. Pattern Anal. Mach. Intell. 12 (1990) 103- 108.

[28] M. Turk, A. Pentland, Eigenfaces for recognition, J. Cogni-tive Neurosci 3 (1) (1991) 71-86.

[29] M. Turk, A. Pentland, Eigenfaces for recognition, J. Cognitive Neurosci. 3 (1) (1991) 71–86.

[30] M.S. Kamel, H.C. Shen, A.K.C. Wong, System for therecognition of human faces, IBM Systems J. 32 (2) (1993)307- 320.

[31] P. J. Philips, P. Grother, R. J. Micheals, D. M. Blackburn, E. Tabassi, J. M. Bone,

[32] P.C. Yuen, D.Q. Dai, G.C. Feng, Wavelet-based PCA for human face recognition, Proceeding of IEEE Southwest Symposium on Image Analysis and Interpretation, 1998,pp. 223-228.

[33] Q. Yang, X. Ding, Symmetrical PAC in face recognition, ICIP02 V (2) (2002) 97–100.

[34] R. Brunelli, T. Poggio, Face recognition: feature verustemplates, IEEE Trans. Pattern Anal. Mach. Intell. 15 (10) (1993) 1042}1052.

[35] S. Prabhakar, S. Pankanti, A. K. Jain, "Biometric Recognition: Security and Privacy

[36] S.C. Chen, D.Q. Zhang, Z.-H. Zhou, Enhanced (PC)2A for face recognition with one training image per person, Pattern Recognition Lett. 25 (10) (2004) 1173–1181.

[37] T. Kanade, Picture processing by computer and recogni-tion of human faces, Technical Report, Tokyo University, 1973.

[38] Trans. on Circuits and Systems for Video Technology, Vol. 14, No. 1, pp 4-19, January 2004.

[39] W. Zhao, A. Krishnaswamy, R. Chellappa, D.L. Swets,J. Weng, Discriminant analysis of principal components for face recognition, Face Recognition: From Theory to Applications, Springer, Berlin , 1988, pp. 73-85.

[40] W. Zhao, R. Chellappa, P.J. Phillips, A. Rosenfeld, Face recognition:a literature survey, ACM Comput. Surv. December Issue (2003) 399–458.

[41] X. Xie, R. Sudhakar, H. Zhuang, On improving eye feature extraction using deformable templates, Pattern Recogni-tion 27 (6) (1994) 791}799.

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